RESEARCH ETHICS APPLICATION - Tea Sense

Project Title

Investigating the Feasibility of Near-Infrared Spectroscopy and AI-Based Predictive Modelling for Rapid Tea Moisture Estimation

Does the research involve human participants?

No.

The research does not involve interviews, surveys, questionnaires, focus groups, or other forms of human participant data collection.

Background to the Research

Tea quality assessment is important to the tea industry, with parameters such as moisture content commonly requiring laboratory-based testing. Conventional moisture measurement methods, including oven drying, can require significant time before a final measurement is obtained.

Recent research has demonstrated the potential of Near-Infrared (NIR) spectroscopy as a rapid and non-destructive technique for analysing tea and other agricultural products. When combined with chemometric and predictive modelling techniques, spectral measurements may be used to estimate selected physical or chemical characteristics.

This research investigates the feasibility of using NIR spectroscopy and predictive modelling to estimate moisture content in selected tea samples. Moisture values obtained through an established oven-drying method will be used as reference measurements. The study will investigate spectral preprocessing, wavelength selection and predictive modelling techniques and evaluate their performance against reference values.

The research does not assume that the proposed technology will replace conventional laboratory testing. Instead, it aims to evaluate whether the approach demonstrates sufficient predictive performance and practical feasibility for rapid tea moisture assessment.

Brief Description and Aims of the Study

The aim of this research is to investigate the feasibility of using Near-Infrared (NIR) spectroscopy and predictive modelling for the rapid estimation of moisture content in tea.

Selected tea samples will be analysed using NIR spectroscopy to obtain spectral measurements. Reference moisture values will be obtained using an oven-drying method. The spectral data will then be processed and analysed using appropriate chemometric and predictive modelling techniques.

The research will investigate preprocessing methods such as Standard Normal Variate (SNV) and Savitzky-Golay filtering. Wavelength-selection approaches, including Competitive Adaptive Reweighted Sampling (CARS), may also be investigated. Predictive models, including Partial Least Squares Regression (PLSR) and other suitable machine-learning approaches, will be evaluated.

The performance of the developed models will be assessed using held-out samples and appropriate statistical measures, including the coefficient of determination (R^2) and Root Mean Square Error of Prediction (RMSEP).

The research will also assess the practical feasibility of conducting rapid spectral measurements under selected operational or field conditions.

Research Design and Methodology

The research will use an experimental quantitative approach.

Tea samples will be obtained for the purpose of spectroscopy and moisture analysis. No human participants will be recruited or studied.

Each selected tea sample will undergo spectral measurement using an appropriate spectroscopy system, primarily focusing on Near-Infrared (NIR) spectroscopy. Corresponding moisture reference values will be obtained using an established oven-drying procedure.

The resulting dataset will contain spectral measurements and associated moisture reference values. Spectral preprocessing techniques, including Standard Normal Variate (SNV) and Savitzky-Golay filtering, will be investigated to reduce noise and variation caused by physical sample characteristics.

Feature or wavelength-selection techniques, such as Competitive Adaptive Reweighted Sampling (CARS), may be evaluated to identify informative spectral variables.

Predictive models will then be developed and evaluated using appropriate statistical and machine-learning techniques. Model performance will be assessed against reference moisture values using held-out samples. Evaluation measures may include R^2 and RMSEP.

Selected practical or field measurements may also be conducted to assess the feasibility of using the approach outside strictly controlled laboratory conditions.

No invasive procedures involving humans or animals will be conducted.

RISK AND HAZARD ASSESSMENT

Potential Hazards or Risks

The research presents a low level of ethical risk because it does not involve human participants, personal data, sensitive personal information, or animal testing.

Potential practical risks include:

- Incorrect interpretation of predictive model outputs.
- Reduced model performance when applied to samples outside the training dataset.
- Possible exposure to laboratory or equipment-related hazards during sample analysis.
- Potential misuse of the research findings if predictive estimates are presented as replacements for certified laboratory measurements without appropriate validation.

Measures to Reduce Risks

The following measures will be implemented:

- Spectroscopy equipment and laboratory equipment will be used according to applicable safety procedures.
- Oven-drying procedures will be conducted using appropriate laboratory safety practices.
- Predictive model outputs will be clearly identified as research estimates.
- Model performance will be evaluated against established reference measurements.
- Held-out samples will be used where possible to assess model generalisation.
- The research will not claim that the proposed system can replace certified laboratory testing unless this is supported by sufficient evidence and validation.
- Any limitations identified during the research will be reported.

Risks Balanced Against Benefits

The anticipated risks of the research are low and can be managed through appropriate laboratory and research procedures.

The potential benefit of the research is the generation of evidence regarding whether spectroscopy and predictive modelling could support faster and non-destructive estimation of tea moisture. The findings may contribute to future research into rapid tea quality assessment.

The potential scientific and practical benefits outweigh the limited risks associated with the research.

Stopping Criteria

The research will be paused or stopped if:

- Equipment failure creates an unacceptable safety risk.
- Laboratory procedures cannot be conducted safely.
- The collected data is found to be unreliable or unsuitable for meaningful analysis.
- Significant methodological limitations make further data collection scientifically unjustifiable.
- Any unforeseen ethical or safety issue arises that cannot be adequately managed.

DATA PROTECTION AND SECURITY

Type of Data Collected

The research will primarily collect:

- Tea sample information relevant to the experiment.
- Spectral measurements.
- Reference moisture measurements.
- Processed and derived analytical data.
- Predictive model outputs.

No personal or sensitive personal data is intended to be collected.

Data Storage and Security

Research data will be stored digitally on password-protected systems and secure institutional or approved cloud storage where available.

Access will be restricted to authorised individuals involved in the research. Research datasets will not contain personal participant information.

Data will be retained in accordance with the University's applicable research and data-retention requirements.

ETHICAL DECLARATION

The proposed research primarily involves the experimental analysis of tea samples using spectroscopy and computational modelling. It does not currently involve human participants, interviews, surveys, questionnaires, or the collection of personal or sensitive personal data.

The research will use appropriate laboratory safety procedures and will evaluate predictive outputs against established reference measurements. The findings will be presented responsibly, including relevant limitations regarding model accuracy, sample diversity and practical applicability.

Any substantial changes to the research methodology that introduce human participants, personal data, interviews, surveys, or additional ethical risks will be submitted for appropriate ethical review and approval before such activities begin.